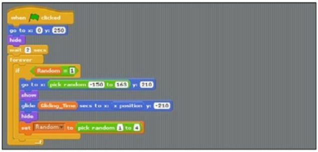

Figure 6: Code for the sprite *obstacle*

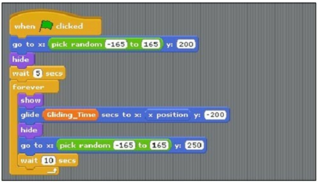

Figure 7: Code for the sprite *fuel*

touches the *tank* sprite. A warning will be displayed when the *fuel* value drops below 15. The game stops as it goes below 1.

Move on to the sprite *gear* and drag down the scripts.

The game starts with the first costume *Gear 1*. Then the value of *gear* and the costume of the sprite *gear* will change according to the gear shifts.

In order to display the countdown, let's include the scripts given in Figure 5 under the sprite *steering wheel*.

When the *Green flag* button is pressed, the countdown will be displayed and the game starts.

The variable *random* stores an integer value between 1 and 4, at random, while executing. The value stored in *random* is used to decide which obstacle has to be displayed on the screen. The speed of the obstacle movement is controlled by the value stored in the variable *gliding_time*, which depends on the gear value.

The code given in Figure 6 is used for all *obstacle*s. Enter values from 1 to 4 for the corresponding obstacle in the *if* block.

Figure 8: Screen shots of the game

Click on the sprite *fuel* and drag down the code given in Figure 7.

These commands enable the sprite *tank* to glide down from random positions with a time interval of 10 seconds.

All the screen shots from the game are given in Figure 8.

The capability to'drag and drop'commands is the key feature of Scratch. The puzzle type arrangement of commands helps the programmer to avoid syntax errors, and the visual constructs provide a better way to understand the control flow in the program. More than just coding, Scratch also provides a better platform for beginners to enhance their creativity. It provides a better, easy to use UI, with which learning becomes more fun and an exciting experience. END

References

[1] *http://www.scratch.mit.edu/about/*
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By: Pranav P

The author is currently studying for a bachelor's degree in Computer Science and Engineering at Amrita University in India. He is an open source enthusiast and also an active member of the Amrita FOSS club. He can be contacted at *pranavp1616@gmail.com.*